

Optimal Hardy Approximation of the Endpoint Logarithm

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Abstract

We compute the exact best polynomial approximation error for the universal endpoint logarithm. Orthogonality makes the Taylor projection optimal, its squared error is the reciprocal-square tail, and the norm is asymptotic to $N^{-1/2}$. The analytic remainder identified in RH-312 is exponentially smaller, so the full deterministic target has the same leading rate. This does not realize the approximants as spectra.

1 Exact projection error

Let $L(w) = \log(1 - w) + w = -\sum_{n \geq 2} w^n/n$, and let $\mathcal{P}_N$ be the polynomials of degree at most N in $H^2(\mathbb{D})$.

Theorem 1.1. *The Taylor section $S_N L = -\sum_{2 \leq n \leq N} w^n/n$ is the unique best approximation to L from $\mathcal{P}_N$, and*

$$E_N(L)^2 := \inf_{p \in \mathcal{P}_N} \|L - p\|_{H^2}^2 = \sum_{n > N} \frac{1}{n^2}.$$

Moreover

$$\frac{1}{N+1} \leq E_N(L)^2 \leq \frac{1}{N}, \quad E_N(L) \sim N^{-1/2}.$$

Proof. The monomials are an orthonormal basis, so the Taylor section is the orthogonal projection. The norm identity is Parseval. Comparing the decreasing function x^{-2} with its upper and lower integrals proves the bounds and the asymptotic. $\square$

2 The complete deterministic target

Write the RH-312 endpoint target as $L + H_{\text{reg}}$, where H_{reg} is analytic on $|w| < r_{\text{reg}}$ for some $r_{\text{reg}} > 1$. Its Taylor tail is $O(r^{-N})$ for every $1 < r < r_{\text{reg}}$. Hence

Corollary 2.1. *The best degree- N approximation error of the full deterministic endpoint target is $N^{-1/2}(1 + o(1))$.*

3 Boundary

If an information class limits usable degree to $N \asymp \log M / \log(q_*/q)$, the theorem yields a $1/\sqrt{\log M}$ approximation scale. This translation is conditional on that degree constraint. It is not a rate theorem for an actual noisy cloud, and Gates A–E remain false/open.